

MFML Mid-Semester Exam — Complete Study Guide

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Compiled from lecture notes, companion notes & 7 years of past papers

 Exam: Saturday

START HERE — Read This First (Step-by-Step)

New to this course? Follow these steps in order before diving into the numbered sections.

Step 1: How to use this file

1. You are reading **MFML_Final_Study_Guide.html** in your browser.
2. Use the **navigation panel on the left** (or top on mobile) to jump to any topic.
3. All formulas render automatically — no other app needed.
4. **How to study one topic:** Read → cover the answer → try the example on paper → check → move on.

Step 2: What is this course?

Mathematics for Machine Learning (MFML) = linear algebra + a bit of calculus, written the way ML uses it.

Idea	Plain English	ML use
$Ax = b$	Solve equations	Find unknown weights / variables
Vector spaces	Sets of vectors with rules	Describe data spaces
Eigenvalues / SVD	Find important directions	PCA, dimensionality reduction
Gradient / Hessian	Slope and curvature	Training / optimization

Study order if starting from zero: Part I → Part II → Part IV → Part V → Part III → Part VII.

Each Part below has its own Starting Guide — read that box first before the numbered sections in that part.

Step 3: What the exam looks like

Every MidSem paper has ~5 questions × ~20 marks:

Question	Topic	Where to learn
Q1	Solve $Ax = b$	Part I · §1–3
Q2	Vector spaces & subspaces	Part II · §4–7
Q3	SVD & eigenvalues	Part IV · §11–13
Q4	Gradient & critical points	Part V · §15
Q5	Inner products & SPD	Part III · §8–9

Part I — Linear Systems & Matrices

Sections 1–3 · Always Q1 on the exam

Starting Guide — Part I

In one sentence: You solve equations by row-reducing a matrix until the answers fall out.

Before you start, know: – A **matrix** is a table of numbers; **$Ax = b$** means “find x that satisfies these equations.” – **Row operations** (swap, scale, add rows) never change the solution. –

REF/RREF = simplified staircase form of the matrix.

What the exam asks (Q1): – Solve $Ax = b \rightarrow$ give **particular + general** solution – Row reduce a matrix $\rightarrow$ find **rank, pivot/free variables, determinant**

Try this first:

Solve $x + 2y = 6$ and $2x + 4y = 12 \rightarrow$ row 2 is $2 \times$ row 1 $\rightarrow$ **infinite solutions**. Let $y = t \rightarrow x = 6 - 2t$.

Answer: $\mathbf{x} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

Rank rule (memorize): - $\text{rank}(A) = \text{rank}([A|b]) = n \rightarrow$ **unique** solution - $\text{rank}(A) = \text{rank}([A|b]) < n \rightarrow$ **infinite** (free variables) - $\text{rank}(A) \neq \text{rank}([A|b]) \rightarrow$ **no** solution

Shortcuts: Delete duplicate rows early · $2 \times 2 \text{ det} = ad - bc$ · inverse exists iff $\text{det} \neq 0$

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1. Systems of Linear Equations

💡 WHAT IS THIS IN PLAIN ENGLISH?

Imagine you have **3 unknowns** (x, y, z) and **3 equations**. A system of linear equations is just a bunch of equations like:

- $x + 2y + z = 10$
- $2x - y + 3z = 5$
- $x + y - z = 2$

$Ax = b$ is just a shorthand for writing all those equations in one go using a matrix. The goal is to **find** x, y, z that satisfy all equations at once.

Think of it like this: each equation is a flat "wall" in space. You're finding where all walls intersect.

The Three Views of $Ax = b$

View	Interpretation
Row view	Each equation = intersection of hyperplanes
Column view	b = linear combination of columns of A
Machine view	Output b produced by input x through matrix A

Three Possible Outcomes

- **Unique solution:** $\text{rank}(A) = \text{rank}([A|b]) = n$ (no. of unknowns) $\rightarrow$ consistent, exactly one solution
- **Infinite solutions:** $\text{rank}(A) = \text{rank}([A|b]) < n \rightarrow$ free variables exist
- **No solution:** $\text{rank}(A) \neq \text{rank}([A|b]) \rightarrow$ inconsistent system

Gaussian Elimination — Step by Step

Elementary Row Operations (preserve solutions): 1. **Swap** two rows: $R_i \leftrightarrow R_j$ 2. **Scale** a row by non-zero scalar: $R_i \rightarrow \lambda \cdot R_i$ ($\lambda \neq 0$) 3. **Eliminate**: $R_i \rightarrow R_i + \lambda \cdot R_j$

Algorithm: 1. Write augmented matrix $[A \mid b]$ 2. Forward elimination $\rightarrow$ Row Echelon Form (REF) 3. Back substitution $\rightarrow$ identify pivot and free variables 4. Further reduce to RREF for cleaner reading

Row Echelon Form (REF) vs Reduced REF (RREF)

REF requirements: 1. All-zero rows at the bottom 2. Each leading entry (pivot) is to the right of the pivot above 3. All entries below a pivot are zero $\rightarrow$ staircase pattern

RREF additional requirements: 4. Each pivot = 1 5. Each pivot is the **only** nonzero in its column

Particular + Homogeneous (General) Solution

Full general solution = Particular solution + Null space

$$\mathbf{x} = \mathbf{x}_{\text{particular}} + \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots$$

where $\mathbf{v}_1, \mathbf{v}_2, \dots$ are basis vectors of the null space (solutions to $A\mathbf{x} = 0$).

Finding the particular solution from RREF: - Set all free variables = 0 - Back-substitute to find pivot variable values

Finding null space basis: - For each free variable, set it = 1 and all others = 0 - Solve for pivot variables $\rightarrow$ one basis vector per free variable

Fully Worked Example — Gaussian Elimination (Exam Style)

FULL STEP-BY-STEP EXAMPLE

Solve: $x + 2y + 3z = 6$, $2x + 4y + 6z = 12$, $x - y = 5$

Step 1 — Write the augmented matrix $[A \mid b]$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & 4 & 6 & 12 \\ 1 & -1 & 0 & 5 \end{array} \right)$$

2 **$R_2 \rightarrow R_2 - 2R_1$** (eliminate x from row 2)

Row 2 becomes $[0, 0, 0 \mid 0] \rightarrow$ this row is all zeros — just disappears!

3 **R3 → R3 – R1** (eliminate x from row 3)

Row 3 becomes $[0, -3, -3 \mid -1]$

4

After elimination the matrix looks like:

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -3 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

2 pivots (columns 1 and 2), 1 free variable (column 3). **rank = 2, nullity = 1.**

5 **Let $z = t$** (free variable — can be anything)

From row 2: $-3y - 3t = -1 \rightarrow y = 1/3 - t$

From row 1: $x + 2(1/3 - t) + 3t = 6 \rightarrow x = 16/3 - t$

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Write the answer:

$$\mathbf{x} = \underbrace{\begin{pmatrix} 16/3 \\ 1/3 \\ 0 \end{pmatrix}}_{\text{particular (set } t=0)} + t \underbrace{\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}}_{\text{null direction}}$$

Shortcut used: Row 2 = 2× Row 1 → spotted immediately, no calculation needed!

⚡ **EXAM TIPS**

- **Before you eliminate:** scan for rows that are multiples of each other. Delete the duplicate instantly.
- **Count free variables** (columns without pivots) = that's how many terms appear in the general solution.
- The **particular solution** is always found by setting all free variables = 0.

⚠ **COMMON MISTAKE**

After getting RREF, students write the eigenvectors of the null space using the **reduced** matrix columns instead of the **original** matrix's pivot columns. Remember: basis of column space = pivot columns of the **original A**, not the reduced one.

2. Determinants

Definition & Geometric Meaning

$\det(A)$ = **signed volume** of the parallelepiped formed by rows (or columns) of A . - $\det(A) \neq 0 \iff A$ is invertible $\iff$ columns/rows are linearly independent

2x2 Matrix

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

3x3 Matrix — Cofactor Expansion along Row 1

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

where cofactor $C_{ij} = (-1)^{i+j}M_{ij}$ and M_{ij} is the (i,j) minor.

Expand along the row/column with the most zeros!

Sarrus' rule (shortcut for 3x3): Copy columns 1,2 beside the matrix, sum diagonals.

Properties of Determinants

Property	Formula
Swap two rows	$\det$ changes sign
Scale a row by λ	$\det$ multiplied by λ
Add multiple of one row to another	$\det$ unchanged
Transpose	$\det(A^T) = \det(A)$
Product	$\det(AB) = \det(A) \cdot \det(B)$
Inverse	$\det(A^{-1}) = 1/\det(A)$
Triangular matrix	$\det$ = product of diagonal entries
Scalar multiple	$\det(cA) = c^n \det(A)$ for $n \times n$ matrix

Triangular / Echelon Form Shortcut

After row reduction to upper triangular matrix U:

$$\det(A) = (-1)^{(\text{number of row swaps})} \times \prod_{i=1}^n u_{ii}$$

Block Diagonal Determinant

$$\det \begin{pmatrix} B & 0 \\ 0 & C \end{pmatrix} = \det(B) \cdot \det(C)$$

Characteristic Equation of cA from A

If char. poly of A is: $\lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_1\lambda + a_0 = 0$

Then char. poly of cA is: $\lambda^n + c \cdot a_{n-1}\lambda^{n-1} + \dots + c^{n-1} \cdot a_1\lambda + c^n \cdot a_0 = 0$

(Multiply coefficient a_k of kth term by $c^{(n-k)}$)

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3. Matrix Inverse

Existence Condition

A^{-1} exists $\iff \det(A) \neq 0 \iff A$ has full rank $\iff A$ is invertible

Finding A^{-1} via Gauss-Jordan

Set up: $[A \mid I]$ and row reduce until left side becomes $I \rightarrow$ right side becomes A^{-1}

$$[A \mid I] \xrightarrow{\text{RREF}} [I \mid A^{-1}]$$

2x2 Formula

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Inverse Properties

Property	Formula
Two-sided	$AA^{-1} = A^{-1}A = I$
Product	$(AB)^{-1} = B^{-1}A^{-1}$
Transpose	$(A^T)^{-1} = (A^{-1})^T$
Scalar	$(cA)^{-1} = (1/c)A^{-1}$

Solving $Ax = b$ using Inverse

If A is invertible: $x = A^{-1}b$ (unique solution)

In exam: often faster to use Gauss elimination directly on $[A|b]$ rather than computing A^{-1} first.

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Part II — Vector Spaces

Sections 4–7 · Always Q2 on the exam

Starting Guide — Part II

In one sentence: A vector space is a set where you can add vectors and scale them; a subspace is a smaller set that still follows the same rules.

Before you start, know: - **Vector space** = set with $+$ and scalar multiplication (examples: $\mathbb{R}^n$, matrices, polynomials) - **Subspace** = subset that is still a vector space — test with **3 steps**: $0 \in W$, closed under $+$, closed under $\cdot$ - **Span** = all linear combinations of given vectors - **Basis** = smallest independent set that spans the space; **dimension** = number of basis vectors

What the exam asks (Q2): - Prove W **is / isn't** a subspace (3-step test) - Find **basis and dimension** of null space or span - Check if a vector **belongs to** W

Try this first:

Is $W = \{(x, y) : x + y = 0\}$ a subspace? **Yes** — $(0,0)$ works, sum of two such pairs still sums to 0, scaling preserves $x + y = 0$.

Is $W = \{(x, y) : x + y = 5\}$ a subspace? **No** — $(0,0) \notin W$ (fail step 1 immediately).

Shortcuts: Null space & span are **always** subspaces (cite theorem) · dim = number of **free variables** after row reduction · n vectors in $\mathbb{R}^n$ dependent $\iff \det = 0$

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4. Vector Spaces & Subspaces

💡 WHAT IS A VECTOR SPACE? (PLAIN ENGLISH)

A **vector space** is a collection of objects (vectors) where two things always work: 1. You can **add** any two objects and stay inside the collection. 2. You can **multiply** any object by a number and stay inside.

Everyday analogy: Think of all arrows starting from the origin. Adding two arrows = tip-to-tail; scaling = stretching. You never leave "arrow space." That's $\mathbb{R}^2$.

A **subspace** is a *smaller* collection that follows the same two rules AND contains the zero vector.

Why does this matter? Solutions to $Ax = 0$ always form a subspace (the null space). Span of any set of vectors is always a subspace. These facts save time in proofs!

8 Axioms of a Vector Space

A set V with operations $(+)$ and $(\cdot)$ is a vector space if: 1. Closure under addition: $u + v \in V$ 2.

Commutativity: $u + v = v + u$ 3. Associativity: $(u + v) + w = u + (v + w)$ 4. Zero vector: $\exists \mathbf{0}$ such that $u + \mathbf{0} = u$ 5. Additive inverse: $\exists -u$ such that $u + (-u) = \mathbf{0}$ 6. Closure under scalar mult: $c \cdot u \in V$ 7.

Distributivity: $c(u + v) = cu + cv$ and $(c+d)u = cu + du$ 8. Identity: $1 \cdot u = u$

Examples of vector spaces: $\mathbb{R}^n$, $m \times n$ matrices, polynomials P_n (dim = $n+1$), continuous functions on $[a,b]$

The 3-Step Subspace Test

$W \subseteq V$ is a subspace $\iff$ ALL THREE hold:

Step	Test
1. Zero vector	$\mathbf{0} \in W$? (plug in zero — if fails, STOP)

Step	Test
2. Closed under +	$u, v \in W \Rightarrow u + v \in W?$
3. Closed under $\cdot$	$c \in \mathbb{R}, u \in W \Rightarrow c \cdot u \in W?$

If all 3 pass $\rightarrow W$ is a subspace (all 8 axioms inherited for free from parent V).

WATCH OUT — Common Failures

Set	Why it FAILS
$\{(x,y) : x + y = 5\}$	Doesn't contain zero vector ($0+0 \neq 5$)
$\{(x,y) : x \geq 0\}$	Fails scalar mult (scaling by -1 exits the set)
$\{(x,y) : xy = 0\}$	Fails addition: $(1,0) + (0,1) = (1,1) \rightarrow 1 \cdot 1 \neq 0$

PASSES — Common Subspaces

Set	Why it PASSES
$W = \{Ax = 0\}$	Null space is always a subspace
$W = \text{span}\{v_1, \dots, v_k\}$	Span of any set is always a subspace
$W = \{(x,y,z) : x+y+z=0\}$	Linear homogeneous equation $\rightarrow$ subspace
Symmetric matrices	Closed under + and scalar mult

Fully Worked Subspace Test (Exam Style)

EXAMPLE — IS W A SUBSPACE?

Q: Is $W = \{(x, y, z) \in \mathbb{R}^3 : 2x - y + z = 0\}$ a subspace?

- Zero vector test:** Does $(0, 0, 0) \in W$?

Check: $2(0) - 0 + 0 = 0 \checkmark$ Zero vector passes.

- Closure under addition:** Take any $u, v \in W$.

$u = (x_1, y_1, z_1)$ with $2x_1 - y_1 + z_1 = 0$ and $v = (x_2, y_2, z_2)$ with $2x_2 - y_2 + z_2 = 0$

$u + v = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$

Check: $2(x_1 + x_2) - (y_1 + y_2) + (z_1 + z_2) = (2x_1 - y_1 + z_1) + (2x_2 - y_2 + z_2) = 0 + 0 = 0 \checkmark$

3 Closure under scalar multiplication: Take $c \in \mathbb{R}$, $u \in W$.

$$cu = (cx, cy, cz)$$

$$\text{Check: } 2(cx) - (cy) + (cz) = c(2x - y + z) = c \cdot 0 = 0 \quad \checkmark$$

✓ **Conclusion:** All 3 steps pass $\rightarrow W$ is a **subspace** of $\mathbb{R}^3$.

Bonus — find its basis: Express $y = 2x + z$, so any vector in W is:

$$(x, 2x+z, z) = x(1, 2, 0) + z(0, 1, 1) \rightarrow \text{Basis} = \{(1, 2, 0), (0, 1, 1)\}, \text{dim} = 2$$

12 34 EXAMPLE — W IS NOT A SUBSPACE

Q: Is $W = \{(x, y) : x + y = 5\}$ a subspace?

1 Zero vector test: Does $(0, 0) \in W$?

$$\text{Check: } 0 + 0 = 0 \neq 5 \times \quad \text{FAIL! Stop here.}$$

Conclusion: W is NOT a subspace. (One failure is enough — no need to check the other two.)

⚡ EXAM SHORTCUT

If W is defined by a **homogeneous equation** (right side = 0, like $2x - y + z = 0$), it is always a subspace — you can state that directly. If the right side is **non-zero** (like $x + y = 5$), fail step 1 immediately.

Subspace of Vector Space of Matrices

Set of all 2×2 symmetric matrices = subspace of $M_{\{2 \times 2\}}$, with **dim = 3**:

$$\text{Basis: } \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

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5. Linear Combinations, Independence & Span

Linear Combination

$\mathbf{v}$ is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_k$ if:

$$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k \quad \text{for some } c_i \in \mathbb{R}$$

To express $\mathbf{v}$ as a linear combination: Solve the system $[\mathbf{v}_1 \mid \mathbf{v}_2 \mid \dots \mid \mathbf{v}_k \mid \mathbf{v}]$ for the coefficients.

Span

$$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \{c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k : c_i \in \mathbb{R}\}$$

Span is automatically a subspace.

Linear Independence

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are **linearly independent** if:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k = \mathbf{0} \implies c_1 = c_2 = \dots = c_k = 0$$

(Only the trivial solution exists)

Vectors are **linearly dependent** if there exists a **non-trivial** combination equal to zero.

Algorithmic Test for Independence (REF Method)

1. Form matrix A with vectors as **columns**
2. Row reduce to REF
3. **Count pivot columns** = number of independent vectors
4. If all columns are pivot columns $\rightarrow$ **independent**
5. Non-pivot columns can be expressed as combinations of pivot columns to their left

Determinant Shortcut (n vectors in $\mathbb{R}^n$)

$$\det(A) \neq 0 \iff \text{vectors are linearly independent}$$

$$\det(A) = 0 \iff \text{vectors are linearly dependent}$$

Checking if Vector is in Span

To check if $\mathbf{u} \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$: - Form augmented matrix $[\mathbf{v}_1 \mid \dots \mid \mathbf{v}_k \mid \mathbf{u}]$ and row reduce - If **consistent** (no $[0 \dots 0 \mid \text{nonzero}]$ row) $\rightarrow$ $\mathbf{u}$ is in span - If **inconsistent** $\rightarrow$ $\mathbf{u}$ is NOT in span

Geometric shortcut: Check if $\mathbf{u}$ satisfies the plane equation $\mathbf{n} \cdot \mathbf{u} = 0$ where $\mathbf{n} = \mathbf{v}_1 \times \mathbf{v}_2$

Polynomials — Independence Test

For $p_1(x), p_2(x), p_3(x) \in P_2$: Set $a \cdot p_1 + b \cdot p_2 + c \cdot p_3 = 0$, collect by powers of x , set each coefficient = 0 $\rightarrow$ solve system.

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6. Basis & Dimension

Definitions

Basis of V = set $B = \{b_1, \dots, b_n\}$ that is: 1. **Linearly independent** (no redundancy) 2. **Spans** V (covers everything)

Every vector in V has a **unique** representation as linear combination of basis vectors.

Dimension $\dim(V)$ = number of vectors in any basis (same for all bases).

Standard Dimensions

Space	Dimension
$\mathbb{R}^n$	n
P_n (polynomials of degree $\leq n$)	$n+1$
$M_{\{m \times n\}}$ ($m \times n$ matrices)	$m \cdot n$
Symmetric $n \times n$ matrices	$n(n+1)/2$

Key Shortcuts

In $\mathbb{R}^n$: any n linearly independent vectors automatically form a basis — no need to re-verify spanning!

Counting trick: Subspace defined by k independent linear equations in $\mathbb{R}^n$ has dimension = $n - k$

Finding Basis from Spanning Set (REF Method)

1. Write spanning vectors as **columns** of matrix A
2. Row reduce to REF

3. Pivot columns of the original matrix A (not the reduced version!) form the basis

Finding Basis from Constraint Equations

For $W = \{(x,y,z) \in \mathbb{R}^3 : 2x - y + z = 0\}$: 1. Express dependent variable: $y = 2x + z$ 2. Write general element: $(x, 2x+z, z) = x(1,2,0) + z(0,1,1)$ 3. Basis = $\{(1,2,0), (0,1,1)\}$, $\dim(W) = 2$

Rank-Nullity Theorem

$$\text{rank}(A) + \text{nullity}(A) = n \quad (\text{number of columns})$$

- $\text{rank}(A) = \text{number of pivot columns} = \dim(\text{column space})$
- $\text{nullity}(A) = \text{number of free variables} = \dim(\text{null space})$

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7. Null Space & Column Space

Null Space (Kernel)

$$\text{Null}(A) = \{x \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$$

Always a subspace. Finding basis: row reduce $A \rightarrow$ identify free variables $\rightarrow$ back-substitute each one.

Column Space (Image/Range)

$$\text{Col}(A) = \text{span of columns of } A = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}$$

Always a subspace. Finding basis: pivot columns of the **original** A .

Row Space

Span of rows of A . Basis = non-zero rows of REF of A .

Key Fact

The null space and column space of A are **perpendicular** in the following sense:

$$\text{rank}(A) + \text{nullity}(A) = n$$

Part III — Inner Products & Orthogonality

Sections 8–10 · Always Q5 on the exam

Starting Guide — Part III

In one sentence: An inner product measures length and angle between vectors; SPD matrices define valid “weighted” inner products.

Before you start, know: - **Dot product:** $\langle x, y \rangle = x^T y$ — gives length $\|x\| = \sqrt{x^T x}$ and angle -

General inner product: $\langle x, y \rangle_A = x^T A y$ where A is **symmetric positive definite (SPD)** -

Orthogonal: $\langle x, y \rangle = 0$ (perpendicular) - **Gram-Schmidt:** turns messy vectors into orthogonal ones with same span

What the exam asks (Q5): - Prove $\langle X, Y \rangle = X^T A Y$ is a valid inner product (symmetry + positive definiteness) - Compute **distance** $d(u, v) = \sqrt{(u - v)^T A (u - v)}$ - Check if two vectors are **orthogonal** under $\langle \cdot, \cdot \rangle_A$

Try this first:

For $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$, distance from $u = (1, 0)$ to $v = (0, 1)$:

$w = u - v = (1, -1) \rightarrow w^T A w = 2 + 3 = 5 \rightarrow \text{distance} = \sqrt{5}$

Shortcuts: Prove SPD via **Sylvester** (leading minors > 0) or **complete the square** · Symmetry needs $A = A^T$

8. Inner Products & Analytic Geometry

WHAT IS AN INNER PRODUCT? (PLAIN ENGLISH)

An **inner product** is a way to measure *how aligned* two vectors are. The standard version is the **dot product**: multiply matching components, then add.

Example: $\langle (1,0), (0,1) \rangle = 0 \rightarrow$ they're perpendicular (orthogonal).

Example: $\langle (1,0), (1,0) \rangle = 1 \rightarrow$ perfectly aligned.

The exam uses **generalised inner products**: $\langle x, y \rangle_A = x^T A y$ where A is an SPD matrix. This "warps" space — distances and angles are measured differently, but all the rules still hold.

Q5 typically asks: Prove this $\langle \cdot, \cdot \rangle$ is a valid inner product (just check symmetry + positive definiteness), then compute a distance or check orthogonality.

Norms

Manhattan (L_1) norm: $\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|$

Euclidean (L_2) norm: $\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2} = \sqrt{\mathbf{x}^T \mathbf{x}}$

Properties: Absolutely homogeneous, Triangle inequality, Positive definite.

Inner Products

Standard dot product: $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$

General inner product with SPD matrix A:

$$\langle \mathbf{x}, \mathbf{y} \rangle_A = \mathbf{x}^T A \mathbf{y}$$

Requirements for valid inner product (bilinear mapping Ω): 1. **Symmetric:** $\Omega(x,y) = \Omega(y,x)$ 2.

Positive definite: $\Omega(x,x) > 0$ for all $x \neq 0$, and $\Omega(0,0) = 0$

Induced Norm

$$\|\mathbf{x}\|_A = \sqrt{\mathbf{x}^T A \mathbf{x}} = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle_A}$$

Distance

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|_A = \sqrt{(\mathbf{u} - \mathbf{v})^T A (\mathbf{u} - \mathbf{v})}$$

Steps to compute distance: 1. Compute $w = u - v$ 2. Compute Aw 3. Compute $w^T(Aw) \rightarrow$ this is $\|w\|_A^2$ 4. Distance = $\sqrt{w^T A w}$

Angles & Orthogonality

Cauchy-Schwarz inequality: $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|$

Angle between vectors:

$$\cos(\omega) = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \cdot \|\mathbf{y}\|}$$

Orthogonality: $\mathbf{x} \perp \mathbf{y} \iff \langle \mathbf{x}, \mathbf{y} \rangle = 0$

Note: vectors orthogonal w.r.t. one inner product may NOT be orthogonal w.r.t. another!

Worked Example — Inner Product Proof + Distance

EXAMPLE: PROVE $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{A} \mathbf{y}$ IS VALID, COMPUTE DISTANCE

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $u = (1, 0)$, $v = (0, 1)$.

1 **Prove symmetry:** $\langle y, x \rangle_A = y^T A x = (y^T A x)^T = x^T A^T y = x^T A y = \langle x, y \rangle_A \checkmark$ (uses $A = A^T$)

2 **Prove positive definite via Sylvester:**

$$D_1 = 2 > 0 \checkmark$$

$$D_2 = \det(A) = 2 \cdot 1 - 1 \cdot 1 = 1 > 0 \checkmark$$

$\rightarrow A$ is SPD $\rightarrow$ inner product is valid $\checkmark$

3 **Compute distance $d(u, v)$:**

$$w = u - v = (1, 0) - (0, 1) = (1, -1)$$

$$Aw = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$w^T (Aw) = (1, -1) \cdot (1, 0) = 1$$

$$d(u, v) = \sqrt{1} = 1$$

Orthogonal Matrix

A square matrix Q is orthogonal $\iff$ its columns are orthonormal:

$$Q^T Q = Q Q^T = I \implies Q^{-1} = Q^T$$

Properties of orthogonal matrices: - Preserve lengths: $\|Qx\| = \|x\|$ - Preserve angles: $\text{angle}(Qx, Qy) = \text{angle}(x, y)$ - $\det(Q) = \pm 1$

Example: 2D rotation matrix $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

9. Symmetric Positive Definite (SPD) Matrices

Definition

A symmetric matrix A is **positive definite** (PD) if:

$$\mathbf{x}^T A \mathbf{x} > 0 \quad \forall \mathbf{x} \neq \mathbf{0}$$

Three Ways to Prove PD

Method 1: Sylvester's Criterion (Leading Principal Minors) A is PD $\iff$ ALL leading principal minors $D_k > 0$: - $D_1 = a_{11} > 0$ - $D_2 = \det(\text{top-left } 2 \times 2) > 0$ - $D_3 = \det(\text{top-left } 3 \times 3) > 0$ - ... - $D_n = \det(A) > 0$

Method 2: Direct Expansion (Completing the Square) Expand $\mathbf{x}^T A \mathbf{x}$ and show it equals a sum of squares ≥ 0 , with equality only at $\mathbf{x} = \mathbf{0}$.

Example: For $A = \begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix}$:

$$Q = 3x^2 - 2xy + y^2 = 2x^2 + (x - y)^2 > 0 \text{ for } (x, y) \neq (0, 0)$$

Method 3: Eigenvalue Check A is PD $\iff$ all eigenvalues of A are **strictly positive** ($\lambda_i > 0$).

Positive Semi-Definite (PSD)

$\mathbf{x}^T A \mathbf{x} \geq 0$ (allows zero for $\mathbf{x} \neq \mathbf{0}$). Eigenvalues ≥ 0 .

Key Facts About SPD Matrices

- SPD matrix has full rank (no non-trivial null space)
- All diagonal entries > 0 (plug in $\mathbf{e}_i$)
- $A^T A$ is always PSD; it is PD if and only if A has full column rank
- Cholesky decomposition $A = LL^T$ exists $\iff A$ is SPD

Block Diagonal SPD Check

$$A = \begin{pmatrix} B & 0 \\ 0 & C \end{pmatrix} \text{ is PD } \iff B \text{ is PD AND } C \text{ is PD}$$

10. Gram-Schmidt Orthogonalization

Goal

Convert a basis $\{v_1, v_2, \dots, v_n\}$ into an **orthonormal basis** $\{q_1, q_2, \dots, q_n\}$.

Classical Gram-Schmidt Process

Step 1: $u_1 = v_1 \rightarrow q_1 = u_1/\|u_1\|$

Step 2: $u_2 = v_2 - \text{proj}_{\{u_1\}}(v_2) = v_2 - (\langle v_2, u_1 \rangle / \langle u_1, u_1 \rangle) u_1 \rightarrow q_2 = u_2/\|u_2\|$

Step 3: $u_3 = v_3 - \text{proj}_{\{u_1\}}(v_3) - \text{proj}_{\{u_2\}}(v_3) \rightarrow q_3 = u_3/\|u_3\|$

General: $\mathbf{u}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \frac{\langle \mathbf{v}_k, \mathbf{u}_j \rangle}{\langle \mathbf{u}_j, \mathbf{u}_j \rangle} \mathbf{u}_j$

BITS Pilani Method (via Gaussian Elimination)

For vectors a, b, c as columns of A :

1. Form augmented matrix $[A^T A \mid A^T]$
2. Apply Gaussian elimination
3. The rows on the right side become mutually orthogonal
4. Normalize to get orthonormal basis

Why it works: $A^T A$ is PD when A has full column rank, so elimination can proceed without row swaps.

Worked Example

Convert $\{(3,1), (2,2)\}$ to orthonormal basis in $\mathbb{R}^2$:

Form $[A^T A \mid A^T] = [[10, 8 \mid 3, 1], [8, 8 \mid 2, 2]]$

After elimination: $[[1, 0.8 \mid 0.3, 0.1], [0, 1 \mid -0.25, 0.75]]$

Rows on right: orthogonal pair. Normalize for orthonormal basis.

QR Decomposition Connection

Gram-Schmidt on columns of A gives: $\mathbf{A} = \mathbf{QR}$ - $\mathbf{Q}$ = matrix with orthonormal columns - $\mathbf{R}$ = upper triangular matrix

Part IV — Eigenvalues & Decompositions

Sections 11–14 · Always Q3 on the exam

Starting Guide — Part IV

In one sentence: Eigenvalues tell you how a matrix stretches certain directions; SVD breaks any matrix into simple rotation-scaling-rotation pieces.

Before you start, know: - **Eigenvalue/eigenvector:** $Av = \lambda v$ — matrix only scales v , doesn't rotate it - Find λ : solve **$\det(A - \lambda I) = 0$** ; find v : solve **$(A - \lambda I)v = 0$** - **2×2 shortcut:**

$\lambda_1 + \lambda_2 = \text{trace}$, $\lambda_1 \lambda_2 = \det$ - **SVD:** $A = U\Sigma V^T$ — compute via eigenvalues of **$A^T A$**

What the exam asks (Q3): - Full **SVD** computation $A = U\Sigma V^T$ - Is matrix **diagonalizable?** (compare algebraic vs geometric multiplicity) - Find eigenvalues/eigenvectors; diagonalize if possible

Try this first:

$A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$ (triangular) → eigenvalues are **3 and 2** (diagonal entries).

For 2×2: trace = 5, det = 6 → $\lambda^2 - 5\lambda + 6 = 0 \rightarrow \lambda = 2, 3 \checkmark$

SVD steps: (1) $A^T A \rightarrow$ (2) eigenvalues → $\sigma_i = \sqrt{\lambda_i} \rightarrow$ (3) $V = \text{eigenvectors} \rightarrow$ (4) $u_i = Av_i/\sigma_i$

Shortcuts: Triangular matrix → eigenvalues on diagonal · Symmetric → always diagonalizable · Orthogonal columns of $A \rightarrow \sigma_i = \|\text{column}_i\|$

11. Eigenvalues & Eigenvectors

WHAT IS AN EIGENVALUE? (PLAIN ENGLISH)

When you multiply a matrix A by most vectors, the result **changes direction**. But there are special "magic" vectors that only get **stretched** (or flipped) without changing direction. Those are **eigenvectors**.

The **eigenvalue** λ tells you *how much* the stretch is: - $\lambda = 2 \rightarrow$ vector doubles in length - $\lambda = -1 \rightarrow$ vector flips direction - $\lambda = 0 \rightarrow$ vector collapses to zero

Why do we care? Eigenvalues tell you the most important directions in a matrix (used in PCA, Google PageRank, quantum mechanics, image compression — everywhere).

Definition

For $n \times n$ matrix A :

$$A\mathbf{x} = \lambda\mathbf{x}, \quad \mathbf{x} \neq \mathbf{0}$$

λ = eigenvalue, $\mathbf{x}$ = eigenvector. Geometrically: A only **stretches** (or flips) the eigenvector direction.

Finding Eigenvalues: Characteristic Equation

$$\det(A - \lambda I) = 0$$

This gives a degree- n polynomial in λ (characteristic polynomial).

Shortcuts for 2x2 Matrix

$$\lambda^2 - (\text{trace}(A))\lambda + \det(A) = 0$$

- Sum of eigenvalues = $\text{trace}(A) = a_{11} + a_{22}$
- Product of eigenvalues = $\det(A)$

Shortcuts for 3x3 Matrix

- Sum of eigenvalues = $\text{trace}(A)$
- Product of eigenvalues = $\det(A)$
- **Triangular/diagonal matrix:** eigenvalues are the diagonal entries (directly!)

Finding Eigenvectors

For each eigenvalue λ : solve $(A - \lambda I)\mathbf{x} = \mathbf{0} \rightarrow$ null space of $(A - \lambda I)$.

Complete Worked Example — 2x2 Eigenvalues & Eigenvectors



FULL EXAMPLE: FIND EIGENVALUES AND EIGENVECTORS

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

1 **Set up $\det(A - \lambda I) = 0$**

$$A - \lambda I = \begin{pmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{pmatrix}$$

$$\det = (4 - \lambda)(3 - \lambda) - (1)(2) = \lambda^2 - 7\lambda + 10 = 0$$

2 **Solve the characteristic equation**

$$\lambda^2 - 7\lambda + 10 = 0 \implies (\lambda - 5)(\lambda - 2) = 0$$

Eigenvalues: $\lambda_1 = 5, \lambda_2 = 2$

Quick check: trace = $4+3 = 7 = 5+2 \checkmark$ $\det = 12-2 = 10 = 5 \times 2 \checkmark$

3 **Find eigenvector for $\lambda_1 = 5$**

Solve $(A - 5I)x = 0$:

$$\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \xrightarrow{R2+2R1} \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix}$$

Free variable: $x_2 = t \rightarrow x_1 = t \rightarrow \mathbf{v}_1 = (1, 1)$

4 **Find eigenvector for $\lambda_2 = 2$**

Solve $(A - 2I)x = 0$:

$$\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \xrightarrow{R2-R1} \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix}$$

$x_2 = t \rightarrow x_1 = -t/2 \rightarrow \mathbf{v}_2 = (-1, 2)$ (scale by 2)

✓ **Verify diagonalization:** Two distinct eigenvalues $\rightarrow$ two independent eigenvectors $\rightarrow A$ is diagonalizable.

$$A = PDP^{-1}, \quad P = \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$$

Algebraic vs Geometric Multiplicity

Term	Definition
Algebraic Multiplicity (AM)	Multiplicity of λ as root of char. polynomial

Term	Definition
Geometric Multiplicity (GM)	$\dim(\text{null space of } (A - \lambda I)) = \# \text{ free variables}$

Always: $1 \leq \text{GM} \leq \text{AM}$

Diagonalizable $\iff \text{GM} = \text{AM}$ for every eigenvalue (i.e., n independent eigenvectors exist)

Checking if λ is an Eigenvalue (Exam Pattern)

Form $(A - \lambda I)$ and check if $\det = 0$, OR row-reduce $(A - \lambda I)$ and check if null space is non-trivial.

If $\text{GM} \geq 1 \rightarrow \lambda$ IS an eigenvalue. Find eigenvector from null space.

Eigenvalues of Special Matrices

Matrix type	Eigenvalues
Triangular	Diagonal entries
Symmetric	All real
Orthogonal	± 1 (complex:
PD matrix	All > 0
PSD matrix	All ≥ 0
Nilpotent ($A^k=0$)	All $= 0$
Idempotent ($A^2=A$)	Only 0 or 1

Using Trace and Determinant to Find Missing Eigenvalue

If one eigenvalue λ_1 is known and matrix is $n \times n$: $-\lambda_1 + \lambda_2 + \dots + \lambda_n = \text{trace}(A)$ - $\lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n = \det(A)$

Use these to solve for unknowns a, b in the matrix.

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12. Matrix Diagonalization & Spectral Theorem

Diagonalization

A is **diagonalizable** if $A = PDP^{-1}$ where: - D = diagonal matrix of eigenvalues - P = matrix of eigenvectors (as columns)

Test: Diagonalizable $\iff$ n linearly independent eigenvectors $\iff GM = AM$ for all eigenvalues.

When Diagonalization FAILS — Nilpotent Matrices

A non-zero matrix A is **nilpotent** if $A^k = 0$ for some k.

Properties of nilpotent A: - All eigenvalues = 0 - A itself $\neq 0$ but $A^k = 0$ - NOT diagonalizable (unless A = 0) - AM > GM for $\lambda = 0$

Classic example:

$$A = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} \implies A^2 = 0$$

- Eigenvalue 0 with AM=2, GM=1 $\rightarrow$ NOT diagonalizable

Spectral Theorem (for Symmetric Matrices)

If A is symmetric ($A^T = A$), then: 1. All eigenvalues are **real** 2. Eigenvectors corresponding to distinct eigenvalues are **orthogonal** 3. A can be orthogonally diagonalized: $A = Q\Lambda Q^T$ - Q = orthogonal matrix of eigenvectors ($Q^T Q = I$) - Λ = diagonal matrix of eigenvalues 4. GM = AM for every eigenvalue (always fully diagonalizable)

Spectral Decomposition

$$A = \lambda_1 \mathbf{q}_1 \mathbf{q}_1^T + \lambda_2 \mathbf{q}_2 \mathbf{q}_2^T + \cdots + \lambda_n \mathbf{q}_n \mathbf{q}_n^T$$

Each term $\lambda_i \mathbf{q}_i \mathbf{q}_i^T$ is a rank-1 matrix.

Maximum of $\mathbf{x}^T \mathbf{A} \mathbf{x}$ Subject to $\|\mathbf{x}\| = 1$

For **any** matrix A (not necessarily symmetric): 1. Find symmetric matrix $B = (A + A^T)/2$ ($\mathbf{x}^T \mathbf{A} \mathbf{x} = \mathbf{x}^T \mathbf{B} \mathbf{x}$ since the antisymmetric part vanishes) 2. $\max \mathbf{x}^T \mathbf{A} \mathbf{x} = \max \mathbf{x}^T \mathbf{B} \mathbf{x}$ = **largest eigenvalue of B** 3. $\min \mathbf{x}^T \mathbf{A} \mathbf{x}$ = smallest eigenvalue of B

Powers of Diagonal Matrices

$$A^k = P D^k P^{-1}, \quad D^k = \text{diag}(\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k)$$

13. Singular Value Decomposition (SVD)

💡 WHAT IS SVD IN PLAIN ENGLISH?

Every matrix A can be thought of as doing three things in sequence: 1. **Rotate** the input (V^T : rotate in input space) 2. **Stretch** along each axis (Σ : scale by singular values) 3. **Rotate** the output (U : rotate in output space)

The σ_i values (singular values) tell you *how much stretching happens* in each direction. The biggest σ_1 = most important direction.

Where is it used? Image compression (keep only top-k singular values), recommender systems (Netflix), noise reduction, data science dimensionality reduction.

Definition

Any $m \times n$ matrix A can be written as:

$$A = U\Sigma V^T$$

Factor	Size	Properties
U	$m \times m$	Orthogonal (left singular vectors)
Σ	$m \times n$	Diagonal-like; $\sigma_i \geq 0$ on diagonal
V	$n \times n$	Orthogonal (right singular vectors)

Singular values: $\sigma_i = \sqrt{\lambda_i(A^T A)}$, arranged in **decreasing order** on Σ diagonal.

The 7-Step SVD Algorithm

Step 1: Compute $A^T A$ ($n \times n$ matrix)

Step 2: Find eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$ of $A^T A$ - Use characteristic equation: $\lambda^2 - \text{trace} \cdot \lambda + \det = 0$ (for 2×2)

Step 3: Compute singular values $\sigma_i = \sqrt{\lambda_i}$

Step 4: Build Σ (same size as A) — put σ_i on diagonal, decreasing order

Step 5: Find eigenvectors of $A^T A \rightarrow$ normalize $\rightarrow$ columns of V - For each λ_i : solve $(A^T A - \lambda_i I)x = 0$, normalize

Step 6: Compute left singular vectors via $\hat{u}_i = \frac{1}{\sigma_i} A \hat{v}_i$

Step 7: If $\sigma_i = 0$ for some i , find remaining u_i via orthogonality conditions

Write: $A = U \Sigma V^T$ ✓

Complete Worked SVD Example

FULL SVD FOR A 2x2 MATRIX

$$A = \begin{pmatrix} 3 & 0 \\ 4 & 0 \end{pmatrix}$$

1 Compute $A^T A$

$$A^T A = \begin{pmatrix} 3 & 4 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 4 & 0 \end{pmatrix} = \begin{pmatrix} 25 & 0 \\ 0 & 0 \end{pmatrix}$$

2 Eigenvalues of $A^T A$

$A^T A$ is diagonal $\rightarrow$ eigenvalues are the diagonal entries: $\lambda_1 = 25$, $\lambda_2 = 0$

3 Singular values $\sigma_i = \sqrt{\lambda_i}$

$$\sigma_1 = \sqrt{25} = 5, \quad \sigma_2 = \sqrt{0} = 0$$

$$\Sigma = \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix}$$

4 V = eigenvectors of $A^T A$ (as columns)

For $\lambda_1=25$: $(A^T A - 25I)v=0 \rightarrow v_1 = (1,0)$

For $\lambda_2=0$: $(A^T A - 0I)v=0 \rightarrow v_2 = (0,1)$

$$V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

5 U columns: $u_i = (1/\sigma_i) \cdot A v_i$

$$u_1 = (1/5) \cdot A \cdot (1,0)^T = (1/5) \cdot (3,4)^T = (3/5, 4/5)$$

u_2 must be orthogonal to u_1 in $\mathbb{R}^2$: $u_2 = (-4/5, 3/5)$

$$U = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}$$

✓ **Final result:**

$$A = U\Sigma V^T = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Verify: $u_1 \sigma_1 v_1^T = (3/5, 4/5) \cdot 5 \cdot (1, 0) = (3, 4, \dots)^T \cdot (1, 0) \rightarrow$ first column of $A = (3, 4) \checkmark$

Verification Using AA^T

Alternatively, find U directly from eigendecomposition of AA^T ($m \times m$), then V from $V^T A = \Sigma^T U$.

Column Orthogonality Shortcut

If columns of A are already orthogonal: - $\|\text{column } j\| = \sigma_j$ directly - $u_j = \text{column } j / \sigma_j$

Example: $A = [[5, 0], [5, 0], [0, -10]]$ - $C_1 \cdot C_2 = 5 \cdot 0 + 5 \cdot 0 + 0 \cdot (-10) = 0 \rightarrow$ orthogonal! - $\sigma_1 = \|C_2\| = \sqrt{0+0+100} = 10$, $u_1 = C_2/10 = [0, 0, -1]^T$ - $\sigma_2 = \|C_1\| = \sqrt{25+25} = 5\sqrt{2}$, $u_2 = C_1/(5\sqrt{2})$

Spectral Norm

$$\|A\|_2 = \sigma_{\max} = \text{largest singular value}$$

$$\max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \sigma_{\max}$$

Low-Rank Approximation (Eckart-Young Theorem)

Best rank- r approximation to A :

$$A_r = \sum_{i=1}^r \sigma_i u_i v_i^T$$

Error: $\|A - A_r\|_2 = \sigma_{r+1}$ (next singular value)

SVD vs Eigendecomposition

	SVD	Eigendecomposition
Applies to	Any matrix	Square matrices only
$A = ?$	$U\Sigma V^T$	PDP^{-1}
U, V	Orthogonal	Generally not

	SVD	Eigendecomposition
Values on Σ	Singular values $\sigma_i \geq 0$	Eigenvalues (can be negative)

Checking Given SVD Components

From exam pattern: given partial SVD $A = U\Sigma V^T$ with some unknown entries (e, λ, γ): 1. Use orthonormality of U columns: $u_i \cdot u_j = 0$ for $i \neq j \rightarrow$ find e 2. Multiply out $A = U\Sigma V^T$, compare with known A entries $\rightarrow$ find λ, γ

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14. Matrix Decompositions: LU, Cholesky, QR

LU Decomposition

$A = LU$ where L = lower triangular, U = upper triangular.

Gaussian elimination steps $\rightarrow$ **L contains the multipliers** (below diagonal), **U is the echelon form.**

Usage: Efficient solving of $Ax = b$ (solve $Ly = b$ forward, then $Ux = y$ backward).

Cholesky Decomposition

$A = LL^T$ (or $A = R^T R$) where L is lower triangular with **positive diagonal**.

Existence: $A = LL^T \iff A$ is symmetric positive definite (SPD).

Cholesky Algorithm (Step by Step)

For $A = LL^T$, compute L column by column:

Diagonal entries:

$$L_{ii} = \sqrt{A_{ii} - \sum_{k=1}^{i-1} L_{ik}^2}$$

Off-diagonal entries ($i > j$):

$$L_{ij} = \frac{1}{L_{jj}} \left(A_{ij} - \sum_{k=1}^{j-1} L_{ik} L_{jk} \right)$$

Cholesky Worked Example

$$A = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

Column 1: - $L_{11} = \sqrt{4} = 2$ - $L_{21} = 2/2 = 1$ - $L_{31} = 2/2 = 1$

Column 2: - $L_{22} = \sqrt{(3 - 1^2)} = \sqrt{2}$ - $L_{32} = (1 - 1 \cdot 1)/\sqrt{2} = 0/\sqrt{2} = 0$

Column 3: - $L_{33} = \sqrt{(3 - 1^2 - 0^2)} = \sqrt{2}$

$$L = \begin{pmatrix} 2 & 0 & 0 \\ 1 & \sqrt{2} & 0 \\ 1 & 0 & \sqrt{2} \end{pmatrix}, \quad A = LL^T \checkmark$$

QR Decomposition

A = QR where Q = orthogonal ($Q^T Q = I$), R = upper triangular.

- Computed via Gram-Schmidt on columns of A
- Q = matrix of orthonormal vectors from Gram-Schmidt
- $R = A^T \cdot Q$ (upper triangular coefficients)

Usage: Least squares, eigenvalue algorithms, solving square systems.

Relationship Between Decompositions

Decomposition	Form	Requirement
LU	$A = LU$	A invertible (with possible row swaps: $PA = LU$)
Cholesky	$A = LL^T$	A symmetric positive definite
QR	$A = QR$	A any $m \times n$ with full column rank
Eigendecomposition	$A = PDP^{-1}$	A square, n independent eigenvectors
Spectral	$A = Q\Lambda Q^T$	A symmetric

Decomposition	Form	Requirement
SVD	$A = U\Sigma V^T$	A any matrix

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Part V — Calculus

Section 15 · Always Q4 on the exam

Starting Guide — Part V

In one sentence: Take partial derivatives to find slope (gradient), then use the Hessian to classify min/max/saddle points.

Before you start, know: - **Partial derivative:** differentiate w.r.t. one variable, treat others as constants - **Gradient** ∇f = vector of all partial derivatives → points **uphill** - **Critical point:** where $\nabla f = 0$ - **Hessian** H = matrix of second partial derivatives → classifies critical points

What the exam asks (Q4): - Compute **gradient** ∇f - Find and **classify critical points** (min / max / saddle) - **Taylor series** expansion up to 2nd order

Try this first:

$$f(x, y) = x^2 + y^2 \rightarrow \frac{\partial f}{\partial x} = 2x, \frac{\partial f}{\partial y} = 2y \rightarrow \nabla f = 0 \text{ at } (0,0).$$

$$\text{Hessian } H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \rightarrow D = 4 > 0, f_{xx} = 2 > 0 \rightarrow \text{local minimum at } (0,0).$$

Classification (2D): $D = f_{xx}f_{yy} - f_{xy}^2$ · $D > 0$ & $f_{xx} > 0 \rightarrow \text{min}$ · $D > 0$ & $f_{xx} < 0 \rightarrow \text{max}$ · $D < 0 \rightarrow \text{saddle}$

Taylor shortcut: substitute $h = x - a$, expand, keep terms up to degree 2

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15. Multivariate Calculus & Optimization

💡 WHAT IS THIS? (PLAIN ENGLISH)

Imagine a hilly landscape where height = $f(x, y)$. You're standing at some point: - **Gradient** ∇f = the direction and steepness of the hill at your feet. Walk in the gradient direction $\rightarrow$ go uphill fastest. - **Critical point** = a flat spot ($\nabla f = 0$). Could be a valley (min), hilltop (max), or a weird saddle shape. - **Hessian** = tells you whether that flat spot is a valley, hilltop, or saddle. - **Taylor series** = a way to approximate the hill near any point using a simple formula.

ML connection: Training a neural network = finding the lowest valley of a "loss landscape." Gradient descent follows $-\nabla f$ downhill.

Partial Derivatives

$$\frac{\partial f}{\partial x_i} = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h}$$

Treat all other variables as constants.

Gradient

$$\nabla f = \begin{pmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \\ \vdots \\ \partial f / \partial x_n \end{pmatrix}$$

The gradient points in the direction of **steepest ascent**.

Hessian Matrix

$$H = \nabla^2 f = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}$$

For a well-behaved function, H is symmetric ($f_{xy} = f_{yx}$).

Critical Points

Set $\nabla f = 0$ and solve the system of equations.

Second Derivative Test — Classification

For a 2D function $f(x, y)$ at critical point (a, b) :

Discriminant: $D = f_{xx} \cdot f_{yy} - (f_{xy})^2 = \det(H)$

Condition	Classification
$D > 0, f_{xx} > 0$	Local Minimum
$D > 0, f_{xx} < 0$	Local Maximum
$D < 0$	Saddle Point
$D = 0$	Inconclusive (test fails)

Alternative: Check eigenvalues of H: - All $\lambda_i > 0 \rightarrow$ Local Minimum - All $\lambda_i < 0 \rightarrow$ Local Maximum - Mixed signs $\rightarrow$ Saddle Point - Any $\lambda_i = 0 \rightarrow$ Inconclusive

Complete Worked Optimization Example (Exam Style)

FULL EXAMPLE: FIND & CLASSIFY ALL CRITICAL POINTS

$$f(x, y) = x^2 + xy + 2y^2 - 4x + 3$$

1 Compute partial derivatives

$$\frac{\partial f}{\partial x} = 2x + y - 4$$

$$\frac{\partial f}{\partial y} = x + 4y$$

2 Set both = 0 and solve

From equation 2: $x = -4y$

Substitute into equation 1: $2(-4y) + y - 4 = 0 \rightarrow -7y = 4 \rightarrow y = -4/7$

Then $x = -4(-4/7) = 16/7$

Critical point: (16/7, -4/7)

3 Compute Hessian matrix

$$H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$$

4 Classify using discriminant $D = \det(H)$

$$D = (2)(4) - (1)^2 = 8 - 1 = 7 > 0$$

$$f_{xx} = 2 > 0$$

$\rightarrow D > 0$ and $f_{xx} > 0 \rightarrow$ **Local MINIMUM at (16/7, -4/7)**

Verification (eigenvalue method):

$$\text{Eigenvalues of H: } \lambda^2 - 6\lambda + 7 = 0 \rightarrow \lambda = 3 \pm \sqrt{2} \approx \{1.59, 4.41\}$$

Both positive $\rightarrow$ H is positive definite $\rightarrow$ **Local minimum confirmed.**

$$f(x,y) = x^2 + xy + 2y^2 - 4x + 3$$

$$\partial f / \partial x = 2x + y - 4 = 0 \quad \partial f / \partial y = x + 4y = 0$$

From second equation: $x = -4y$ Substitute: $2(-4y) + y = 4 \rightarrow -7y = 4 \rightarrow y = -4/7$ $x = 16/7$

Hessian: $H = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$ $D = 2 \cdot 4 - 1^2 = 7 > 0$, $f_{xx} = 2 > 0 \rightarrow$ **Local Minimum**

Eigenvalue check: $\lambda^2 - 6\lambda + 7 = 0 \rightarrow \lambda = 3 \pm \sqrt{2}$ (both > 0) $\rightarrow$ **PD $\rightarrow$ Local Minimum** 

Taylor Series

1D (around $x = a$):

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \dots$$

Maclaurin series ($a = 0$):

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

2D Taylor expansion around (a,b) :

$$f(x,y) \approx f(a,b) + f_x(a,b)(x - a) + f_y(a,b)(y - b) + \frac{1}{2} \begin{pmatrix} x - a \\ y - b \end{pmatrix}^T H \begin{pmatrix} x - a \\ y - b \end{pmatrix}$$

Exam Pattern: Taylor Polynomial via Substitution (Faster!)

Q: Find degree-2 Taylor polynomial of $g(x) = f(x, 1)$ at $x = 1$

Method: Set $h = x - 1$, substitute $x = 1 + h$ into $g(x)$, expand and keep only h^0, h^1, h^2 terms.

Example: $g(x) = x^2 - 3x + 5$ - Set $h = x - 1$: $g(1+h) = (1+h)^2 - 3(1+h) + 5 = 1 + 2h + h^2 - 3 - 3h + 5 = 3 - h + h^2$ - Taylor: $g(x) = 3 - (x-1) + (x-1)^2$

Composition example: $f(x) = \exp(\sin(x^2))$ at $x = 0$

- $\sin(x^2) \approx x^2$ (for small x , ignore x^6 terms)
- $e^u \approx 1 + u + u^2/2 \dots$ with $u = x^2$
- $P(x) = 1 + x^2$ (degree-2)

Range of $\exp(\sin(x^2))$: $\sin$ can be -1 to 1 , so range = $[e^{-1}, e^1] = [1/e, e]$

Chain Rule

Scalar: $\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = (\nabla f)^T \frac{d\mathbf{x}}{dt}$

Vector (Jacobian): If $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $\mathbf{x} = \mathbf{g}(t)$, then:

$$\frac{d\mathbf{f}}{dt} = J_f \cdot \frac{d\mathbf{x}}{dt}$$

where $J_f = \begin{pmatrix} \partial f_1 / \partial x_1 & \cdots & \partial f_1 / \partial x_n \\ \vdots & & \vdots \\ \partial f_m / \partial x_1 & \cdots & \partial f_m / \partial x_n \end{pmatrix}$

Matrix Gradient

For $K = A A^T$, the gradient w.r.t. A :

$$\frac{\partial K}{\partial A} = 2A$$

Linearization

Near a point $\mathbf{x}_0$, the best linear approximation:

$$L(\mathbf{x}) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T (\mathbf{x} - \mathbf{x}_0)$$

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Part VI — Special Matrices

Section 16 · Sometimes tested

Starting Guide — Part VI

In one sentence: These are special matrix types with fixed properties — know the definitions and how to prove (or disprove) diagonalizability.

Before you start, know: - **Nilpotent:** $A^k = 0$ for some $k \rightarrow$ all eigenvalues = 0, usually **not** diagonalizable - **Householder:** $C = I - 2ww^T \rightarrow$ symmetric, orthogonal ($C^2 = I$), $\det = -1$ -

Companion matrix: built from a polynomial $\rightarrow$ its eigenvalues = roots of that polynomial

What the exam asks: - Show a matrix is **nilpotent** (compute $A^2, A^3...$) - Prove **not diagonalizable** via $AM \neq GM$ for eigenvalue 0 - Verify **Householder** properties (symmetric, orthogonal, $C^2 = I$)

Try this first:

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \rightarrow A^2 = 0 \rightarrow \text{nilpotent of order 2.}$$

Char poly: $\lambda^2 = 0 \rightarrow AM = 2$ for $\lambda = 0$. $\text{Null}(A)$ has dim 1 $\rightarrow GM = 1 \rightarrow$ **not diagonalizable**.

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16. Special Matrix Types

Nilpotent Matrices

- **Definition:** $A^k = 0$ for some positive integer k
- **All eigenvalues are 0** (but eigenvectors may exist)
- **NOT diagonalizable** (unless $A = 0$)
- $AM > GM$ for eigenvalue 0

How to show NOT diagonalizable: 1. Find characteristic polynomial $\rightarrow \lambda^2 = 0 \rightarrow AM = 2$ 2. Find null space of $(A - 0I) = A \rightarrow \text{rank}(A) = 1 \rightarrow \text{nullity} = 1 \rightarrow GM = 1$ 3. $GM \neq AM \rightarrow$ NOT diagonalizable

Alternative: Compute A^2 ; if $A \neq 0$ but $A^2 = 0 \rightarrow$ nilpotent of order 2 $\rightarrow$ not diagonalizable.

Householder / Reflector Matrices

Form: $C = I - 2\mathbf{w}\mathbf{w}^T$ where $\mathbf{w}$ is a unit vector ($\|\mathbf{w}\| = 1$)

Properties: 1. Symmetric: $C^T = (I - 2\mathbf{w}\mathbf{w}^T)^T = I - 2\mathbf{w}\mathbf{w}^T = C$ 2. Orthogonal: $C^TC = CC = I \rightarrow C^2 = I \rightarrow C = C^{-1}$ 3. $\det(C) = -1$ (orientation-reversing) 4. Eigenvalues: -1 (for $\mathbf{w}$ direction) and $+1$ (for all directions $\perp \mathbf{w}$)

Proof of orthogonality:

$$C^TC = (I - 2\mathbf{w}\mathbf{w}^T)(I - 2\mathbf{w}\mathbf{w}^T) = I - 4\mathbf{w}\mathbf{w}^T + 4\mathbf{w}(\mathbf{w}^T\mathbf{w})\mathbf{w}^T = I - 4\mathbf{w}\mathbf{w}^T + 4\mathbf{w}\mathbf{w}^T = I$$

(using $\mathbf{w}^T\mathbf{w} = 1$)

Companion Matrix

For polynomial $p(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_1\lambda + c_0$, the companion matrix is:

$$C = \begin{pmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & -c_{n-1} \end{pmatrix}$$

The eigenvalues of C are precisely the roots of $p(\lambda)$.

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Part VII — Exam Prep

Sections 17–19 · Read last, review often

Starting Guide — Part VII

In one sentence: This part tells you exactly what repeats every year and gives copy-paste proof templates for each question type.

How to use this part: 1. Read **§17 Exam Strategy** — know which topics appear every year and the 5-question structure 2. Print or memorize **§18 Formula Sheet** — use during last-day review 3. Practice **§19 BONUS Templates** — walk through each template once on paper before the exam

Night-before checklist: - ☐ Can you solve $Ax = b$ with particular + general solution? - ☐ Can you do the 3-step subspace test in under 2 minutes? - ☐ Can you compute SVD for a 2×2 or 3×2 matrix? - ☐ Can you find gradient, critical points, and classify with Hessian? - ☐ Can you prove an inner product is valid (symmetry + PD)?

Exam time strategy: Q1 (linear systems) first — it's always there and builds confidence. Save SVD for when you're warmed up.

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17. Exam Strategy & Recurring Question Patterns

Topic Frequency Across 7 Years of Past Papers

Topic	Frequency
Linear Systems (REF, rank, solutions)	🔴 Every exam
Vector Spaces / Subspaces	🔴 Every exam
Eigenvalues & Eigenvectors	🔴 Every exam
SVD (full computation)	🔴 Every exam
Inner Products / SPD matrices	🔴 Every exam
Gradient & Hessian	🔴 Every exam
Taylor Series	🟡 Most exams
Basis & Dimension	🟡 Most exams
Cholesky / LU / QR	🟡 Sometimes
Nilpotent / Householder matrices	🟡 Sometimes

Standard Question Structure (5 Questions, ~20 marks each)

Q1: Linear Systems (always appears) - (a) Solve $Ax = b$ — find particular + general solution - (b) Row reduce matrix B — find echelon form, rank, determinant - Sub-parts: check existence, identify free variables

Q2: Vector Spaces (always appears) - (a) Show W is/isn't a subspace (3-step test) - (b) Find basis and dimension of null space / span - (c) Check if vector $u \in W$ (consistency test) - Variants: prove span is subspace, prove null space is subspace

Q3: SVD or Eigenvalues (always appears, often both) - (a) Full SVD computation $A = U\Sigma V^T$ - (b) Is matrix diagonalizable? (AM vs GM argument) - (c) Given partial SVD $\rightarrow$ find unknowns

Q4: Calculus (always appears) - (a) Compute gradient ∇f - (b) Find and classify critical points (Hessian) - (c) Taylor series expansion - Variant: find critical points of $g(x,y) = f(x,y,0)$

Q5: Inner Products (always appears) - (a) Verify inner product axioms (symmetry + PD) — use Sylvester's or completing square - (b) Compute distance $d(u,v)$ using $\|w\|_A = \sqrt{w^T A w}$ - (c) Verify orthogonality: $\langle x, y \rangle_A = x^T A y = 0$?

Common Exam Tricks & Shortcuts

- 1. Row observation (fast elimination):** Before doing full elimination, check if rows are multiples → discard instantly.
- 2. Trace/det for 2×2 eigenvalues:** $\lambda_1 + \lambda_2 = \text{trace}$, $\lambda_1\lambda_2 = \text{det}$ → often faster than characteristic equation.
- 3. Zero determinant check for dependence:** n vectors in $\mathbb{R}^n$: form matrix, $\text{det} = 0 \iff$ dependent.
- 4. Count free variables for dimension:** $\text{dim} =$ number of free variables after row reduction.
- 5. Spectral theorem shortcut:** Symmetric matrix → $\text{GM} = \text{AM}$ always → just find eigenvectors of distinct values and use orthogonality for repeated ones.
- 6. SVD column orthogonality:** If columns of A are orthogonal → $\sigma_i = \|\text{column}_i\|$ and $u_i = \text{column}_i/\sigma_i$ → no $A^T A$ needed!
- 7. Taylor by substitution (fastest):** Let $h = x - a$, substitute, expand, keep degree ≤ 2 .
- 8. Checking if SPD by block diagonal:** If A is block diagonal, check each block separately.
- 9. Proving null space is subspace:** Standard proof: $0 \in \text{Null}$ ($A \cdot 0 = 0$), $Au=0$ and $Av=0 \rightarrow A(u+v)=0$, $A(cu)=c \cdot Au=0$.
- 10. Finding 1D subspace:** Just pick any single non-zero basis vector from the basis you found.

Time Allocation Strategy (120 minutes)

Question	Suggested Time
Q1: Linear Systems	20 min
Q2: Vector Spaces	25 min
Q3: SVD/Eigenvalues	30 min
Q4: Calculus	25 min
Q5: Inner Products	20 min
Buffer	0 min left

Priority: Read all 5 questions first (2 min). Start with the question you're most confident in. Partial credit is available — show all working steps!

18. Quick-Reference Formula Sheet

Linear Algebra

General solution: $\mathbf{x} = \mathbf{x}_p + \text{Null}(A)$

$\text{rank}(A) + \text{nullity}(A) = n$ (number of columns)

$\det(AB) = \det(A) \det(B)$, $\det(A^T) = \det(A)$, $\det(cA) = c^n \det(A)$

$\det(\text{triangular}) = \prod \text{diagonal entries}$

Eigenvalues

$\det(A - \lambda I) = 0$ (characteristic equation)

$\sum \lambda_i = \text{trace}(A)$, $\prod \lambda_i = \det(A)$

Spectral: $A = Q\Lambda Q^T$ (symmetric A)

SVD

$A = U\Sigma V^T$, $\sigma_i = \sqrt{\lambda_i(A^T A)}$, $\hat{u}_i = \frac{1}{\sigma_i} A\hat{v}_i$

$\|A\|_2 = \sigma_{\max}$, $\|A - A_r\|_2 = \sigma_{r+1}$

Inner Products

$\langle \mathbf{x}, \mathbf{y} \rangle_A = \mathbf{x}^T A \mathbf{y}$, $d(u, v) = \sqrt{(\mathbf{u} - \mathbf{v})^T A (\mathbf{u} - \mathbf{v})}$

$\cos \omega = \frac{\langle x, y \rangle}{\|x\| \|y\|}$, $\perp \iff \langle x, y \rangle = 0$

SPD

A is PD $\iff D_k > 0 \forall k \iff$ all eigenvalues $> 0 \iff A = LL^T$

Calculus

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)^T$$

$$H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}, \quad D = \det(H) = f_{xx}f_{yy} - f_{xy}^2$$

D > 0, f_{xx} > 0	Min	D > 0, f_{xx} < 0	Max	D < 0	Saddle	D = 0	Inconclusive
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$$f(x) \approx f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2$$

$$e^x \approx 1 + x + \frac{x^2}{2}, \quad \sin x \approx x, \quad \cos x \approx 1 - \frac{x^2}{2} \text{ (near 0)}$$

Cholesky

$$L_{ii} = \sqrt{A_{ii} - \sum_{k < i} L_{ik}^2}, \quad L_{ij} = \frac{1}{L_{jj}} \left(A_{ij} - \sum_{k < j} L_{ik} L_{jk} \right)$$

Householder

$$C = I - 2\mathbf{w}\mathbf{w}^T, \quad \|\mathbf{w}\| = 1 \implies C^T = C, \quad C^T C = I, \quad \det(C) = -1$$

Subspace Test (3 Steps)

$$0 \in W ? \quad u, v \in W \Rightarrow u + v \in W ? \quad c \in \mathbb{R}, u \in W \Rightarrow cu \in W ?$$

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BONUS: Worked Templates for Each Exam Question Type

Template A: Proving $V = \text{span}\{v_1, v_2\}$ is a subspace

- 1. **Zero vector:** Let coeff = 0. $0 \cdot v_1 + 0 \cdot v_2 = \mathbf{0} \in V \checkmark$
- 2. **Closure under +:** Let $u = av_1 + av_2$ and $w = bv_1 + bv_2$. Then $u+w = (a+b)v_1 + (a+b)v_2 \in V \checkmark$
- 3. **Closure under $\cdot$:** $k \cdot u = (ka)v_1 + (ka)v_2 \in V \checkmark$

Alternatively: "Span of any set of vectors is always a subspace" (Theorem — state this for full marks)

Template B: Proving Null Space is a Subspace

1. $A\mathbf{0} = \mathbf{0} \rightarrow \mathbf{0} \in \text{Null}(A) \checkmark$
2. $Au = 0, Av = 0 \rightarrow A(u+v) = Au + Av = 0 + 0 = 0 \rightarrow u+v \in \text{Null}(A) \checkmark$
3. $Au = 0 \rightarrow A(cu) = c \cdot Au = c \cdot 0 = 0 \rightarrow cu \in \text{Null}(A) \checkmark$

Template C: Full SVD Walkthrough

Given A ($m \times n$): 1. Compute $A^T A$ 2. $\lambda^2 - \text{trace} \cdot \lambda + \det = 0 \rightarrow \lambda_1, \lambda_2$ 3. $\sigma_1 = \sqrt{\lambda_1}, \sigma_2 = \sqrt{\lambda_2}$ (ordered: $\sigma_1 \geq \sigma_2$) 4. $\Sigma = [[\sigma_1, 0], [0, \sigma_2], [0, 0]]$ (match dimensions of A) 5. For λ_1 : solve $(A^T A - \lambda_1 I)x = 0 \rightarrow$ normalize $\rightarrow v_1$ 6. For λ_2 : solve $(A^T A - \lambda_2 I)x = 0 \rightarrow$ normalize $\rightarrow v_2$ 7. $u_1 = (1/\sigma_1) \cdot Av_1$ (normalize if needed) 8. $u_2 = (1/\sigma_2) \cdot Av_2$; remaining u_i from orthogonality 9. Write: $A = [u_1 | u_2 | \dots] \cdot \Sigma \cdot [v_1 | v_2 | \dots]^T$

Template D: Inner Product Verification

To show $\langle X, Y \rangle = X^T A Y$ is a valid inner product:

Symmetry: $\langle Y, X \rangle = Y^T A X = (Y^T A X)^T = X^T A^T Y = X^T A Y = \langle X, Y \rangle \checkmark$ (since $A = A^T$)

Positive definiteness: Use Sylvester ($D_1 > 0, D_2 > 0$) OR complete the square.

Template E: Finding Basis of Null Space

Given A (row reduce): 1. Row reduce $A \rightarrow \text{RREF}$ 2. Identify pivot columns and free variable columns 3. For each free variable $x_j = 1$ (others = 0): - Back-substitute to find pivot variable values - That vector is one basis vector 4. Basis = {all such vectors}, $\dim$ = number of free variables

Good luck on Saturday! You've covered every topic from 7+ years of past papers. Trust your preparation!

Study Guide compiled from: 8 Companion Notes, 3 Lecture PDFs, 2 Webinars, Cholesky + LU/QR + Particular/Homogeneous notes, 3 Practice sets, 10+ MidSem papers (2020–2026) + Saurabh's full solutions.